Program Design Quiz 3

(Please use C language to solve the problem)

題目敘述：

請撰寫一個多回合的「四位數 1A2B」判別器。系統首先給定回合數 N（正整數）。每一回合開始時，會提供一組目標數字 S；接著讀入多筆猜測 G。對於每一次猜測，請依規格輸出評分結果，直至該回合被猜中為止；然後進入下一回合，直到完成 N 回合。

形式化定義

目標數字 S

- 為長度恰為 4 的十進位數字字串（0－9），允許前導 0。
- 四個數字互不重複。
- 測資保證每回合提供之 S 合法。

猜測 G 與有效性

- 每次猜測讀入一行數字 G。
- 若 G 不是「長度恰為 4 且為正整數數字」的字串，該筆猜測視為無效，輸出 Invalid input!。
  - 範例：-5、27948 等皆為 Invalid input。
- 若 G 合法但含重複數字，不因此判無效（測資保證數字不重複）。

評分（xAyB）

當 G 為合法四位數字字串且 G != S 時，輸出 xAyB，其中：

- A（位置命中數）：

$$(A = \{\, i \in \{0,1,2,3\} \mid G[i] = S[i] \,\})$$

- B（數字命中但不同位之配對數）：
  先移除所有已計入 A 的同位字元，再就「剩餘的 S 與 G」分別統計各數字 d ∈ {0,…,9} 的出現次數，取逐字元次數的最小值總和：

$$[B = \sum_{d=0}^{9} \min\{\mathrm{count}_{\mathrm{S_rem}}(d),\ count_{\mathrm{G_rem}}(d)\}]$$

  亦即 B 嚴格遵循「配對不重複」原則，只計數字正確但位置不同的匹配量。

成功條件

- 當且僅當 G = S（四個字元完全相同）時，輸出 Success，並結束當前回合，進入下一回合。
- 評測保證每回合最終會出現至少一次 G = S。

輸入說明：

第一行為正整數 **M**，表示回合數。

接續每一回合（共 M 回合）：

- 第 1 行：字串 S（本回合之目標數字；合法且四位互異，允許前導 0）。
- 第 2 行起：多行字串，各為一次猜測 G，直到出現 Success!!的那一筆為止；之後立即進入下一回合的 S。

輸出說明：

自每回合的第一筆猜測起，對每一筆 G 立即輸出一行結果，為下列之一：

1. Invalid input!：若 G 非恰為 4 位的數字字串。
2. Success!!：若 G == S；輸出後結束該回合。
3. xAyB：否則，依上節定義計算 A 與 B，無多餘空白或文字。

Examples

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Sample Input 1:

1
1234
5678
4321
1290
1234

Sample Output 1:

0A0B
0A4B
2A0B
Success!!

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Sample Input 2:

2
4271

1234
4721
-5
4271
4985
4985

Sample Output 2:
1A2B
2A2B
Invalid input!
Success!!
Success!!

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Problem Description：

Implement a **multi-round** judge for the "four-digit 1A2B" game. The system first provides the number of rounds N (a positive integer). At the start of each round, a **target number** S is given; then multiple **guesses** G are read. For each guess, output the result according to the specification until the round is solved; then proceed to the next round, until all N rounds are completed.

Formal Definitions

Target Number S

- A string of exactly 4 decimal digits (0–9), leading zeros allowed.
- The four digits are pairwise distinct.
- The test data guarantees that each round's S is valid.

Guess G and Validity

- Each guess is read as one line representing a guess G.
- If G is **not** a string of **exactly four decimal digits** (digits only, no sign), the guess is **invalid**; output Invalid input!.
  - Examples: -5, 27948 are both **Invalid input!**.
- If G is valid but contains repeated digits, it is not considered invalid (the test data guarantees non-repeated digits).

Scoring (xAyB)

When G is a valid 4-digit string and G != S, output xAyB, where:

- A (position matches):

$$(A = \{\, i \in \{0,1,2,3\} \mid G[i] = S[i] \,\})$$

- B (digit matches in different positions):
  First remove all positions already counted in A. On the remaining characters of S and G, for each digit $d \in \{0,\cdots,9\}$, count its occurrences and sum the per-digit minimums:

$$[B = \sum_{d=0}^{9} \min\{\mathrm{count}_{\mathrm{S_rem}}(d),\ count_{\mathrm{G_rem}}(d)\}]$$

That is, B strictly follows a non-overcounting pairing rule: only digits that are correct but placed in different positions are counted, and no digit is paired more than once.

Success Condition

- If and only if $G=S$ (all four characters identical), output Success, and end the current round, then proceed to the next round.
- The test data guarantees that in every round there will be at least one guess with $G=S$.

Input：

Line 1: A positive integer M, the number of rounds.

For each of the M rounds:

- Line 1: The string S (the target number of the round; valid, 4 distinct digits, leading zeros allowed).
- From Line 2 onward: Multiple lines, each a guess G, until a line that produces Success!!; then immediately proceed to the next round's S.

Output：

Starting from the first guess of each round, for every G output exactly one line, which must be one of:

1. Invalid input!：if G is not exactly a 4-digit string.
2. Success!!：if G == S; after printing this line, end the round.
3. xAyB：otherwise, compute A and B as defined above; print with no extra spaces or text.

Examples

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Sample Input 1:

1
1234
5678
4321
1290
1234

Sample Output 1:

0A0B

0A4B
2A0B
Success!!

==========

Sample Input 2:
2
4271
1234
4721
-5
4271
4985
4985

Sample Output 2:
1A2B
2A2B
Invalid input!
Success!!
Success!!